

2018EBN111C3**VRAESTEL**

Kopiereg voorbehou

Mei 2018

2018EBN111C3**QUESTION PAPER**

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May 2017

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 min + 10 min to complete OCR form <i>30 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		6 + OCR	

Important

- The examination regulations of the University of Pretoria apply.
- All questions must be answered on the Optical Character Recognition (OCR) form.
- Read the instructions on the OCR form carefully.
- Answers must include units, and must be written as rational numbers.

Dosent(e) / <i>Lecturer(s):</i>	D le Roux, D Ramotsoela, X Ye
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Do step 0 in the first 5 minutes of the session. (0-5 min)**Do step 1 in the first 30 minutes of the test. (5-35 min)****Do step 2 within the following 10 minutes. (35-45 min)****STEP 0 (-5 min to 0 min):** Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN111C3**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

- Complex numbers:Rectangular form: always place the “j” **at the start** of the imaginary part of the complex number.

Polar form: always add the “ ° ” (degree sign) after the angle, and make a clear “ ∠ ” (angle) sign.

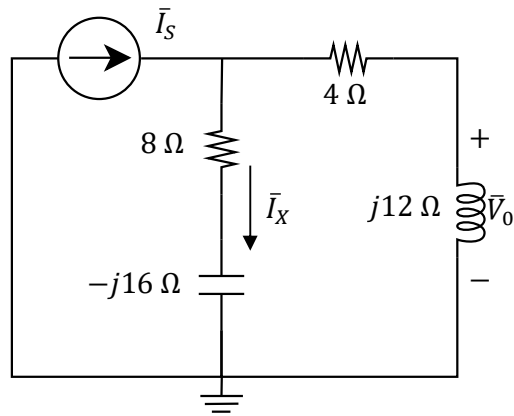
Examples

01	–	1	.	4	+	j	8			μ	A						
02	3	4	0	∠	4	5	°			V							
03	–	j	5	4						k	Ω						
04	3	3	0	0						r	a	d	÷	s			
05	5	0	∠	–	1	3	0	°		m	A						

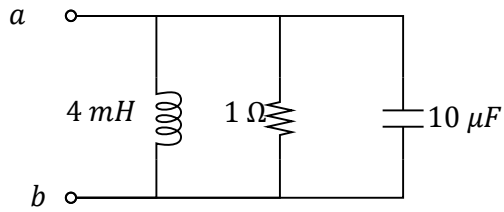
Questions 01-02 [2]		Vraag 01-02 [2]	
The following sinusoid is given: $v_1 = 16 \sin(\omega t + 130^\circ) \text{ V}$ It is known that - v_2 leads v_1 by 160°. - v_1 lags v_3 by 80°.		Die volgende sinusied word gegee: $v_1 = 16 \sin(\omega t + 130^\circ) \text{ V}$ Dit is bekend dat: - v_2 loop voor v_1 met 160°. - v_1 loop na v_3 met 80°.	
01	What is the phase angle ϕ_d between v_2 and v_3 ? ($0 \leq \phi_d \leq 180^\circ$)	01	Wat is die fasehoek ϕ_d tussen v_2 en v_3 ? ($0 \leq \phi_d \leq 180^\circ$)
02	Choose the correct statement: 1 v_2 leads v_3 2 v_3 leads v_2 (Write 1 if statement 1 is correct, and 2 if statement 2 is correct.)	02	Kies die korrekte stelling: 1 v_2 loop voor v_3 2 v_3 loop voor v_2 (Skryf 1 indien stelling 1 reg is, en 2 indien stelling 2 reg is.)

Question 03 [2]		Vraag 03 [2]	
The following three elements are all connected in series: a $5 \cos(3000t) \text{ V}$ voltage source, a $20 \text{ } \Omega$ resistor, and a 5 mF inductor. What is the voltage across the inductor? (Give the answer in polar format. Assume the positive form of Ohm's Law.)		Die volgende drie elemente word in series gekoppel: 'n $5 \cos(3000t) \text{ V}$ spanningsbron, 'n $20 \text{ } \Omega$ resistor, en 'n 5 mH induktor. Wat is die spanning oor die induktor? (Give the answer in polêre formaat. Aanvaar die positiewe vorm van Ohm se Wet.)	
03	$\bar{V}_L$	03	$\bar{V}_L$

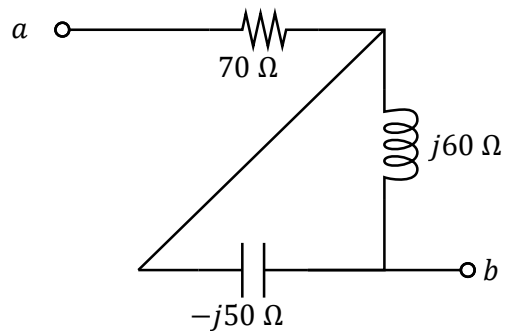
Questions 04-05 [4]		Vraag 04-05 [4]	
Given: $\bar{I}_X = 5\text{ A}$. Determine $\bar{V}_0$ and $\bar{I}_S$.		Gegee: $\bar{I}_X = 5\text{ A}$. Bepaal $\bar{V}_0$ en $\bar{I}_S$.	
04	$\bar{V}_0$ (Rectangular form)	04	$\bar{V}_0$ (Rechthoekige vorm)
05	$\bar{I}_S$ (Rectangular form)	05	$\bar{I}_S$ (Rechthoekige vorm)



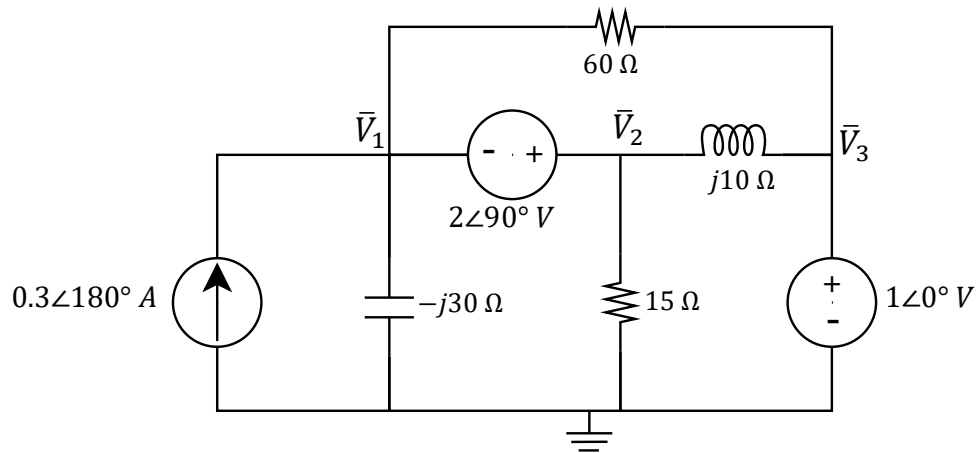
Question 06 [2]		Vraag 06 [2]	
06	For what value of ω will Z_{ab} be equal to: $Z_{ab} = 1 + j0 \Omega$ (Enter the unit as: $rad \div s$ on the OCR form.)	06	Vir watter waarde van ω sal Z_{ab} gelyk wees aan: $Z_{ab} = 1 + j0 \Omega$ (Skryf die eenheid as: $rad \div s$ op die OKH vorm.)



Question 07 [2]		Vraag 07 [2]	
07	Calculate Z_{ab} . (Rectangular format)	07	Bereken Z_{ab} . (Reghoekige formaat)



Question 08-11 [4]				Vraag 09-12 [4]			
Use nodal analysis to set up two equations of the following form:				Use nodal analysis to set up two equations of the following form:			
$a\bar{V}_1 + b\bar{V}_2 = 6 - j17$				$a\bar{V}_1 + b\bar{V}_2 = 6 - j17$			
$c\bar{V}_1 + d\bar{V}_2 = j2$				$c\bar{V}_1 + d\bar{V}_2 = j2$			
Give the coefficients in rectangular form .				Give the coefficients in rectangular form .			
08	a	09	b	08	a	09	b
10	c	11	d	10	c	11	d



PRACTICE PAGE / OEFENBLAD

Assessment ID

2	0	1	8	E	B	N	1	1	1	C	3		
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Remember:

Write a complex number in:

- rectangular form as $a \pm j b$ where j is at the **start** of the imaginary number.
- polar form as $a \angle \theta^\circ$ where the angle sign $\angle$ is clear, and you add the degree unit after the angle).

(See examples on page 1.)

	Answer	Unit
01	$\phi_d =$	
02	Statement #:	No unit required
03	$\bar{V}_L =$ (polar)	
04	$\bar{V}_0 =$ (rectangular)	
05	$\bar{I}_S =$ (rectangular)	
06	$\omega =$	rad ÷ s
07	$Z_{ab} =$ (rectangular)	
08	$a =$ (rectangular)	No unit required
09	$b =$ (rectangular)	
10	$c =$ (rectangular)	
11	$d =$ (rectangular)	